

Global Electric Vehicle (EV) Adoption & Market Trends Visualization

Course: CIS 671 – Information Visualization

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1. Introduction

Electric Vehicles (EVs) are transforming global transportation systems and driving the shift toward sustainable mobility. Our project, *Global Electric Vehicle Adoption & Market Trends Visualization*, aims to explore how EV adoption has grown globally, how different powertrain types are distributed, and what regions contribute most to this transformation.

We designed an interactive dashboard using D3.js and Observable to answer key questions relevant to policymakers, researchers, and sustainability advocates, using real-world data from 2010–2023.

2. Dataset Overview

We used the IEA Global EV Dataset (2010–2023) from Kaggle, which contains over 5,800 records. After cleaning and filtering, we retained the following attributes:

- year: From 2010 to 2023
- region: Geographic location (e.g., USA, China, Europe)
- category: Vehicle type (Cars, Buses, Trucks, Vans)
- parameter: Measurement metric (EV sales, EV stock, Oil displacement, Electricity demand)
- powertrain: Technology type (BEV, PHEV, FCEV)
- mode: Transport usage mode (Private, Public, Freight)
- unit: Unit of measurement (e.g., Thousand, GWh)
- value: Numeric value corresponding to the above context

The dataset was preprocessed in Python (Colab) to remove missing values, filter relevant parameters, and convert types. We then grouped and formatted data into separate CSV for D3 visualizations.

3. Visualization Design & Tools Used

We used D3.js within Observable notebook to build six diverse visualizations. These were chosen to represent trends in a time series, comparisons across categories, and geographic distribution.

- **Karthika:** Line Chart, Grouped Bar Chart, Multi-line Chart
- **Devendhar:** Donut Charts, Choropleth Map, Bar Chart

Data preprocessing was done in Python, and styling/custom interaction was handled in JavaScript using D3 libraries.

4. Visualizations & Insights

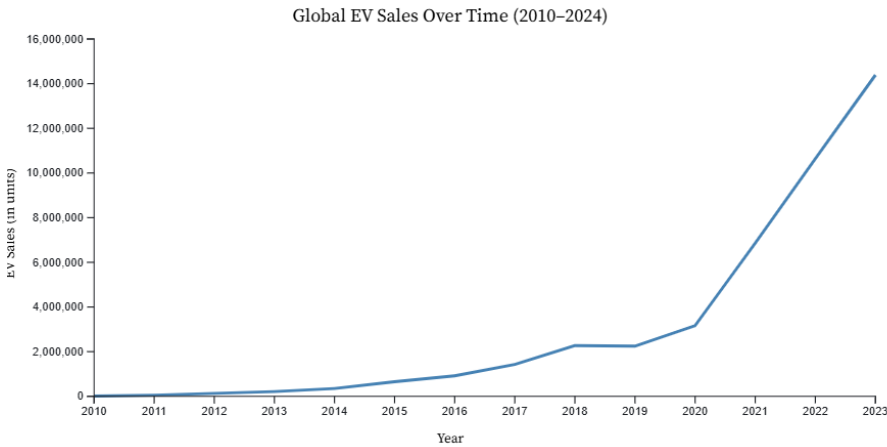
4.1 Line Chart – Global EV Sales Over Time (2010–2024)

Attributes Used: year, parameter = EV Sales, value

This line chart illustrates the trend of global electric vehicle (EV) sales from 2010 to 2024. It shows how the market has evolved over more than a decade, transitioning from a niche segment to a major global industry.

Insight:

EV sales remained relatively low until around 2017, after which there was a noticeable rise. From 2020 onward, sales increased dramatically, crossing 15 million units by 2023. This indicates a global shift in consumer behavior and policy support, especially after the pandemic period, likely fueled by climate initiatives, subsidies, and increased EV model availability.



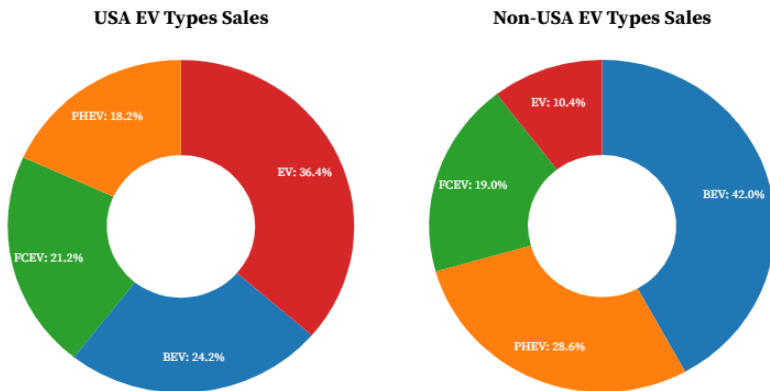
4.2 Donut Charts – Powertrain Distribution: USA vs non-USA

Attributes Used: region, powertrain, value

These side-by-side donut charts compare the distribution of electric vehicle powertrain types between the USA and Non-USA regions. Each slice represents the share of EV types sold, highlighting BEV (Battery Electric Vehicle), PHEV (Plug-in Hybrid Electric Vehicle), FCEV (Fuel Cell Electric Vehicle), and a general EV category.

Insight:

In the USA, BEVs account for 24.2%, PHEVs make up 18.2%, FCEVs hold 21.2%, and general EVs lead with 36.4%. In contrast, non-USA regions are more BEV-dominated, with BEVs at 42.0%, PHEVs at 28.6%, FCEVs at 19.0%, and other EVs contributing only 10.4%. This contrast highlights that while the USA has a more balanced mix of EV technologies, international regions have significantly leaned toward BEV adoption—driven by infrastructure availability and aggressive emission-reduction targets.



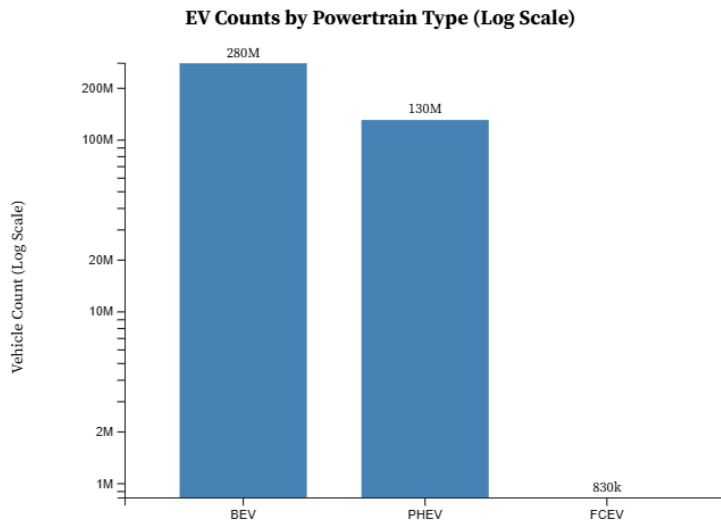
4.3 Bar Chart – EV Counts by Powertrain Type (Log Scale)

Attributes Used: powertrain, value (aggregated counts)

This bar chart compares the total number of electric vehicles globally by powertrain type using a log scale for clarity.

Insight:

BEVs dominate the market with 280 million units, followed by PHEVs at 130 million, and FCEVs trailing far behind at just 830,000. The use of a log scale helps visualize this steep disparity, highlighting BEVs as the industry standard while PHEVs still hold relevance. FCEVs, on the other hand, are yet to achieve significant global adoption.



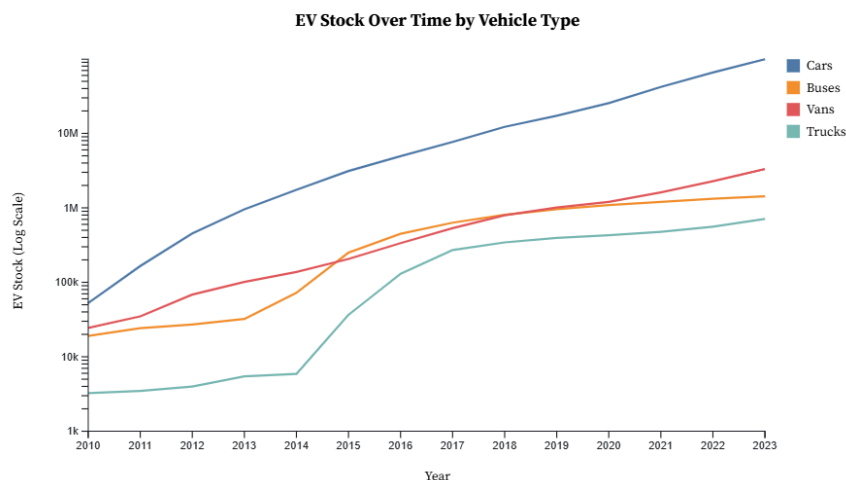
4.4 Multi-line Chart – EV Stock Over Time by Vehicle Type

Attributes Used: year, category, value

This line chart compares how EV stock has grown across four major vehicle types — Cars, Buses, Vans, and Trucks from 2010 to 2023. A log scale was used to clearly visualize growth patterns, especially across categories with varying magnitudes.

Insight:

Cars show the steepest and most consistent growth, crossing 10 million by 2023, establishing them as the primary contributor to global EV stock. Vans and buses follow with steady but moderate increases, while trucks show the slowest growth. This highlights that while commercial electrification is progressing, private passenger EVs are driving the majority of adoption.



4.5 Choropleth Map – EV Stock by Country (2023)

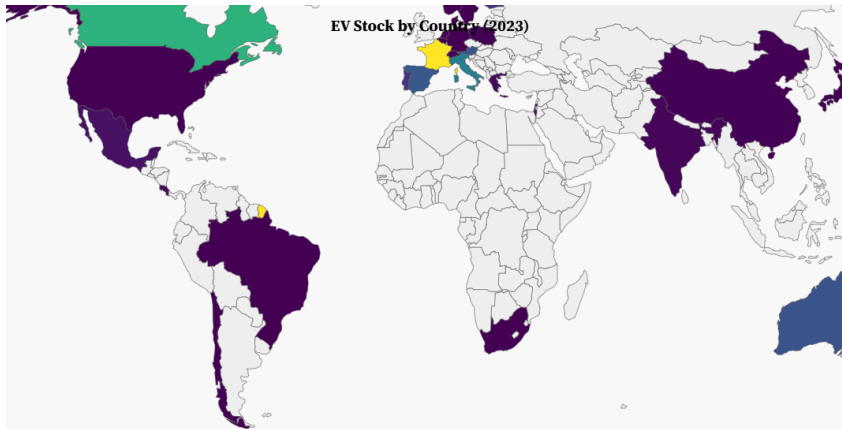
Attributes Used: region, parameter = EV stock, year = 2023, value

This choropleth map displays electric vehicle stock across countries in the year 2023, using color intensity to represent volume.

Insight:

China, the USA, and several European countries (notably France and Germany) have the highest EV stock, indicating strong market presence and infrastructure. In contrast, most countries across Africa, South America, and parts of Southeast Asia

show very low or no visible EV stock, signaling disparities in EV access and adoption. The map emphasizes the global imbalance in EV penetration, with North America, Europe, and East Asia being the primary EV hubs in 2023.



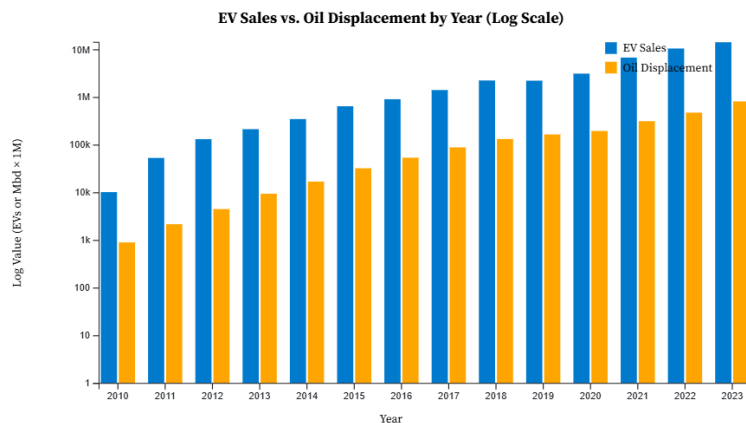
4.6 Grouped Bar Chart – EV Sales vs. Oil Displacement by Year (Log Scale)

Attributes Used: year, parameter = EV Sales & Oil Displacement, value (log scale)

This grouped bar chart compares annual EV sales and the corresponding barrels of oil displaced from 2010 to 2023. A log scale is applied to show the wide range of values effectively.

Insight:

Both EV sales and oil displacement have shown a consistent upward trend, particularly post-2015. While EV sales are consistently higher, oil displacement is steadily catching up, reflecting the time-lagged impact of EV penetration on fossil fuel consumption. The visualization underscores the growing environmental significance of electric vehicles as they become more mainstream.



5. Research Questions Addressed

Our visualizations aimed to answer the following:

- How has global EV adoption evolved over time?
- How Do EV Powertrain Types Differ Between USA and the rest of the world?
- Which Powertrain Types Dominate the EV Landscape Globally?
- Which EV Vehicle Types Are Growing the Fastest Over Time?
- Where Is EV Adoption Highest Around the World?
- Is EV Adoption Effectively Reducing Oil Dependence?

These questions directly shaped the design and narrative of each visualization.

6. Target Audience & Use Cases

This dashboard is designed for:

- Policy Makers: Support evidence-based subsidy and infrastructure planning
- Automotive Industry: Monitor global EV demand trends
- Sustainability Researchers: Study adoption patterns across time and regions
- Educators/Advocates: Promote data-driven awareness campaigns

7. Challenges & Solutions

- Ensuring color consistency across donuts and stacked bar charts
- Aligning units and scaling between EV stock and oil displacement
- Rendering performance in Observable with large datasets
- Avoiding redundancy while ensuring each chart provided unique value

8. Conclusion

This dashboard captures key patterns in global EV adoption, comparing region-wise technology use, vehicle trends, and environmental impact. Our visualization suite helps translate large-scale tabular data into a story that supports clean transportation policy and innovation.

With EVs continuing to rise, such visual tools will be crucial in tracking progress, identifying gaps, and guiding the future of mobility.

9. References

- Kaggle Dataset: Global EV Data
- Observable Notebook: [Observable link](#)